Astronomical Instruments

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

1. Instruments extend the eye
2. Exposure time and signal-to-noise
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1. Instruments extend the eye

Astronomical instruments collect more light, resolve finer detail, separate wavelengths and record time. The main choices are aperture, focal length, detector, filter, mount and calibration. A visual observer, a photometrist and a spectroscopist may choose different instruments for the same target.

The practical question is not “What is the biggest telescope?” but “What measurement do I need, and what noise or systematic error will limit it?”



Bias · dark · flat · standard stars

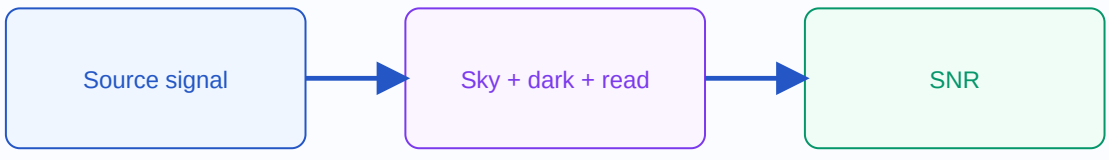
Figure 2. Raw counts are not magnitudes until zero points and extinction are handled.

2. Exposure time and signal-to-noise

For a photon-counting detector, the signal grows with exposure time. Noise comes from source photons, sky background, dark current and read noise. A simplified model, with S, B, D the aperture totals and n_{pix} the number of pixels in the aperture, is

$$\mathrm{SNR} = \frac{S}{\sqrt{S + B + D + n_{\mathrm{pix}}\sigma_{\mathrm{read}}^2}}, \tag{1}$$

If the source dominates, SNR $\propto \sqrt{t}$. If the sky dominates, SNR still grows as $\sqrt{t}$ but more slowly. Read noise matters most for short exposures. Doubling exposure does not double SNR; it increases it by $\sqrt{2}$ in the photon-noise limit.



SNR = S / √(S + B + D + n_pix R²)

Aperture totals; longer exposure helps until systematics dominate.

Figure 1. Bright sky, dark current and read noise set different limits.

3. Filters and photometric systems

Filters define the bandpass. A V filter approximates visual magnitude; B and R give colour information; narrowband filters isolate emission lines. Photometry requires a zero point, extinction correction and often colour terms. Raw counts are not magnitudes.

Differential photometry compares a target with stable comparison stars in the same field. It cancels many transparency and instrumental trends, but it cannot remove differences in colour, airmass or pixel response if the stars are not treated consistently.

4. Spectroscopy and resolution

A spectrograph spreads light by wavelength. The resolving power is

$$R = \frac{\lambda}{\Delta\lambda}. \tag{2}$$

High resolution separates lines and measures velocities; low resolution collects more photons per resolution element and is better for faint sources. The Doppler shift gives velocity: $\Delta\lambda/\lambda = v/c$ for nonrelativistic speeds.

5. Calibration and observing strategy

Bias, dark and flat frames remove instrumental signatures. Standard stars set the zero point. Wavelength lamps or sky lines calibrate spectra. Tracking and guiding prevent trailing. For time-series work, keep exposure time, filter, focus and comparison stars consistent.

Plan exposures before observing: estimate SNR, check saturation, choose cadence, and allow time for calibration. A well-planned small telescope can outperform a poorly planned large one.

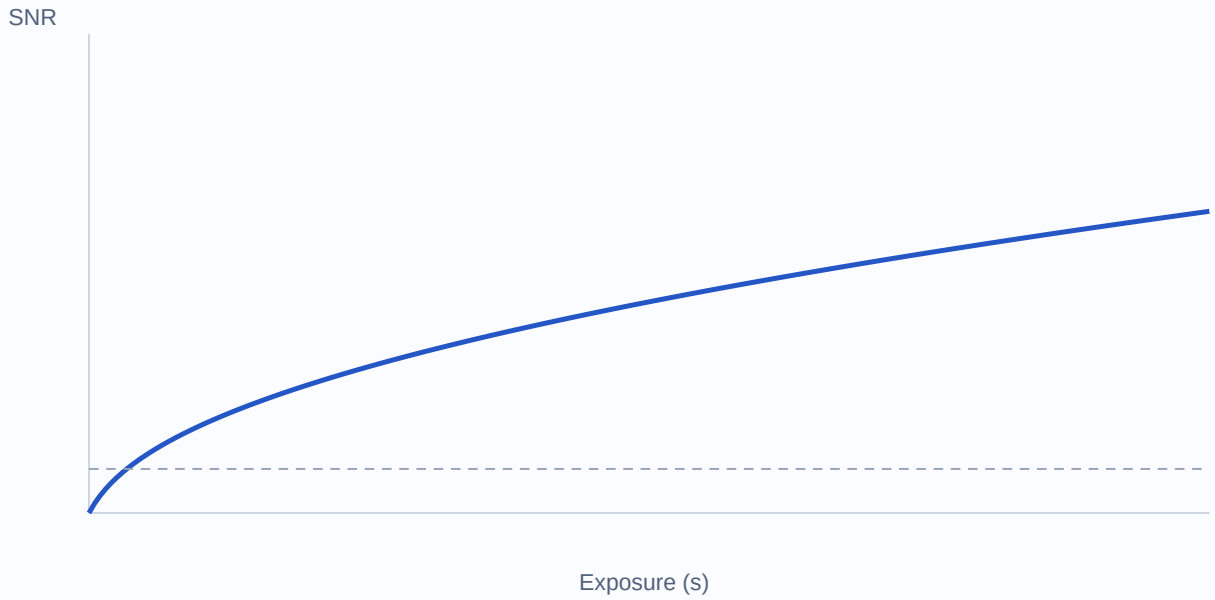
INTERACTIVE EXPERIMENT

Signal, sky and noise in one exposure.

A simplified photon-counting model uses a fixed zero point, 1 arcsec² photometric aperture, read noise and dark current. It is a planning tool, not a calibrated instrument model.

Source electrons	Sky electrons	Signal-to-noise
2.40e+2	**6.03e-2**	**14.72**

Time for SNR = 5
10.1 s



In the source-dominated regime SNR grows as √t; in the sky-dominated regime it also grows as √t but with a smaller coefficient. Read noise matters for short exposures.

Try this. Double the exposure. Does SNR double? Make the sky brighter and find the new exposure needed for SNR = 5.

Snapshot of the experiment. Open the HTML edition to change the controls.

6. Safety and responsibility

Never point an unfiltered telescope at the Sun. Solar observation requires certified front-aperture filters and proper supervision. Lasers, heavy mounts, cold nights and remote sites also require safety planning. Good technique includes protecting the observer as well as the data.

A reliable method. Define the measurement, estimate the photon budget, choose the filter and cadence, calibrate, and record the uncertainty. The instrument is part of the argument.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — SNR estimate. An exposure gives 10,000 source electrons, 2500 sky electrons and 100 dark electrons in the aperture. The total read-noise variance in that aperture is 25 electrons² (equivalently, one pixel with 5-electron read noise). Find the SNR.

Worked solution →

[2] Problem 2 — Exposure for SNR 5. A source produces 100 electrons/s and the sky produces 100 electrons/s in the aperture. Read noise and dark current are negligible. How long is needed for SNR 5?

Worked solution →

[1] Problem 3 — Pixel scale. A camera has 4 µm pixels on a 500 mm focal-length telescope. Find the pixel scale.

Worked solution →

[2] Problem 4 — Spectral resolution. A spectrograph has $R = 5000$ at 500 nm. What is $\Delta\lambda$? What velocity resolution does this correspond to?

Worked solution →

[2] Problem 5 — Doppler shift. A line at 500 nm is shifted by 0.05 nm. Estimate the radial velocity.

Worked solution →

[1] Problem 6 — Calibration purpose. Match bias, dark and flat frames to their main corrections.

Worked solution →

[2] Problem 7 — Differential photometry. Why does comparing a target with a nearby stable comparison star reduce transparency errors? When can it fail?

Worked solution →

[2] Problem 8 — Saturation. A detector saturates at 50,000 electrons per pixel. A star produces 10,000 electrons/s in the brightest pixel. Neglecting sky, dark current and read noise, what is the maximum exposure before saturation? Why might you choose a shorter exposure?

Worked solution →

[3] Problem 9 — Cadence. An eclipse lasts 2 hours. You want at least 20 points during the eclipse and 10 points on each side, with 1 hour of out-of-eclipse baseline on each side. If each exposure is 60 s, what is the minimum total integration time, and what elapsed session length and cadence does the plan require? What overheads might increase the session length?

Worked solution →

[3] Problem 10 — Instrument choice. You want to measure a 0.02 mag eclipse of a magnitude 10 star. Would you prioritize high spectral resolution, large aperture, stable photometry or wide field? Justify your choice and name one systematic to control.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.